R Code

```r
############################################################
# PRACTICAL 5: MOVING AVERAGE (MA) MODELS USING R
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# Objective:
# To understand the structure of Moving Average (MA) models
# and to fit MA models of different orders using R.
#
# The practical also aims to interpret the estimated
# MA coefficients and relate the MA model
# to the dependence structure studied using ACF and PACF.
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# ============================================================
# 1. INTRODUCTION TO THE DATASET
# ============================================================

# We use the built-in AirPassengers dataset.

data(AirPassengers)

# Display the dataset
AirPassengers


# ============================================================
# 2. BASIC INFORMATION ABOUT THE DATA
# ============================================================

# Class of the data
class(AirPassengers)

# Number of observations
length(AirPassengers)

# Starting time
start(AirPassengers)

# Ending time
end(AirPassengers)

# Frequency of the data
frequency(AirPassengers)


# ============================================================
# 3. PLOT THE TIME SERIES
# ============================================================

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```r
plot(AirPassengers,
     main = "Monthly International Airline Passengers",
     xlab = "Year",
     ylab = "Number of Passengers")


# ============================================================
# 4. AUTOCORRELATION FUNCTION (ACF)
# ============================================================

acf(AirPassengers,
    main = "ACF of AirPassengers")


# ============================================================
# 5. PARTIAL AUTOCORRELATION FUNCTION (PACF)
# ============================================================

pacf(AirPassengers,
     main = "PACF of AirPassengers")


# ============================================================
# 6. FITTING AN MA(1) MODEL
# ============================================================

# Fit an exact MA(1) model

ma1 <- arima(AirPassengers,
             order = c(0, 0, 1))

# Display the fitted MA(1) model
ma1


# ============================================================
# 7. EXTRACT THE MA(1) COEFFICIENT
# ============================================================

# Extract the estimated MA coefficient

ma1$coef

# Extract only the MA(1) coefficient

theta1 <- ma1$coef["ma1"]

theta1


# ============================================================
# 8. ESTIMATED MEAN OF THE SERIES
# ============================================================

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```r
# Extract the estimated mean

mu1 <- ma1$coef["intercept"]

mu1


# ============================================================
# 9. INNOVATION VARIANCE
# ============================================================

# Extract the innovation variance

ma1$sigma2


# ============================================================
# 10. OBTAIN THE FITTED MA(1) EQUATION
# ============================================================

# Display estimated values

theta1
mu1


# Display the fitted MA(1) equation

cat("Fitted MA(1) model:\n")

cat("Y(t) =",
    round(mu1, 4),
    "+ epsilon(t) +",
    round(theta1, 4),
    "* epsilon(t-1)\n")


# ============================================================
# 11. FITTING AN MA(2) MODEL
# ============================================================

# Fit an exact MA(2) model

ma2 <- arima(AirPassengers,
             order = c(0, 0, 2))

# Display the fitted MA(2) model

ma2


# ============================================================
# 12. EXTRACT MA(2) COEFFICIENTS
# ============================================================
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# Extract the MA(2) coefficients

ma2$coef

# Extract individual coefficients

theta1_ma2 <- ma2$coef["ma1"]

theta2_ma2 <- ma2$coef["ma2"]

theta1_ma2
theta2_ma2


# ============================================================
# 13. ESTIMATED MEAN FOR MA(2)
# ============================================================

mu2 <- ma2$coef["intercept"]

mu2


# ============================================================
# 14. INNOVATION VARIANCE FOR MA(2)
# ============================================================

ma2$sigma2


# ============================================================
# 15. OBTAIN THE FITTED MA(2) EQUATION
# ============================================================

# Display estimated values

theta1_ma2
theta2_ma2
mu2


# Display the fitted MA(2) equation

cat("Fitted MA(2) model:\n")

cat("Y(t) =",
    round(mu2, 4),
    "+ epsilon(t) +",
    round(theta1_ma2, 4),
    "* epsilon(t-1) +",
    round(theta2_ma2, 4),
    "* epsilon(t-2)\n")

```

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# ============================================================
# 16. FITTING AN MA(3) MODEL
# ============================================================

# Fit an exact MA(3) model

ma3 <- arima(AirPassengers,
             order = c(0, 0, 3))

# Display the fitted MA(3) model

ma3


# ============================================================
# 17. EXTRACT MA(3) COEFFICIENTS
# ============================================================

# Extract the MA(3) coefficients

ma3$coef

# Extract individual coefficients

theta1_ma3 <- ma3$coef["ma1"]

theta2_ma3 <- ma3$coef["ma2"]

theta3_ma3 <- ma3$coef["ma3"]


theta1_ma3
theta2_ma3
theta3_ma3


# ============================================================
# 18. ESTIMATED MEAN FOR MA(3)
# ============================================================

mu3 <- ma3$coef["intercept"]

mu3


# ============================================================
# 19. INNOVATION VARIANCE FOR MA(3)
# ============================================================

ma3$sigma2


# ============================================================
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```r
# 20. OBTAIN THE FITTED MA(3) EQUATION
# ============================================================

# Display estimated values

theta1_ma3
theta2_ma3
theta3_ma3
mu3


# Display the fitted MA(3) equation

cat("Fitted MA(3) model:\n")

cat("Y(t) =",
    round(mu3, 4),
    "+ epsilon(t) +",
    round(theta1_ma3, 4),
    "* epsilon(t-1) +",
    round(theta2_ma3, 4),
    "* epsilon(t-2) +",
    round(theta3_ma3, 4),
    "* epsilon(t-3)\n")


# ============================================================
# 21. COMPARISON OF MA COEFFICIENTS
# ============================================================

# MA(1) coefficient

ma1$coef


# MA(2) coefficients

ma2$coef


# MA(3) coefficients

ma3$coef



################################################################
# PRACTICE QUESTION
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# Dataset: UKgas

# Using the built-in UKgas dataset:
#
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```r
# (a) Load and display the dataset.
#
# (b) Find its class, length, starting time,
#     ending time and frequency.
#
# (c) Plot the time series.
#
# (d) Plot the ACF and PACF.
#
# (e) Fit an MA(1) model.
#
# (f) Extract the estimated MA coefficient.
#
# (g) Extract the estimated mean of the series.
#
# (h) Obtain the fitted MA(1) equation by
#     substituting the estimated parameter values.
#
# (i) Interpret the estimated MA coefficient.
#
################################################################
```